

NagarVoice - Project Submission

Problem Statement Selected

Problem Statement 2: Community Hero - Hyperlocal Problem Solver

Solution Overview

NagarVoice is an AI-powered hyperlocal civic issue reporting platform that enables citizens to identify, report, verify, and track infrastructure problems such as potholes, broken streetlights, water leakages, and waste management issues in their communities.

Citizens upload a photo of an issue, and Google Gemini's Vision API automatically analyzes the image to classify the problem, assess severity, and suggest the responsible municipal department - reducing manual data entry to almost zero. Reports appear instantly on a live, citizen-friendly map where community members can verify issues through upvoting, increasing their priority for resolution.

Beyond reporting, NagarVoice introduces a predictive intelligence layer. By analyzing 30 days of historical issue data across geographic zones, the platform forecasts which areas are likely to experience infrastructure problems next - shifting civic management from reactive to proactive. A real-time City Health Score gives citizens and authorities an instant, quantified view of their city's infrastructure condition, while a Gemini-powered conversational assistant lets users ask natural language questions about civic data in their area.

The platform also addresses two often-overlooked needs in civic tech: anonymous reporting for sensitive issues, and transparent accountability through a public "Still Waiting" board highlighting the longest-unresolved problems in the city.

Key Features

- AI Photo Analysis (Reporter Agent): Gemini Vision automatically categorizes uploaded photos, assigns severity scores, suggests the responsible department, and auto-fills the report form
- Duplicate Detection (Verifier Agent): Automatically checks new reports against nearby existing issues to prevent duplicate entries and merge similar reports
- Smart Prioritization (Prioritizer Agent): Calculates a dynamic priority score for every issue based on severity, community votes, and report age
- Predictive Insights (Predictor Agent): Analyzes 30-day issue history by zone to forecast infrastructure risk and generate

actionable recommendations for authorities

- Citizen-Friendly Live Map: Real-time map showing all reported issues with color-coded status pins, heatmap pulse animations for high-density problem zones, and one-tap location finding
- City Health Score: A real-time, algorithmically calculated 0-100 score reflecting overall civic infrastructure health
- Community Verification: Citizens can verify and upvote issues reported by others, increasing report credibility and priority
- Gamification: Points and badges system rewarding citizens for reporting and verifying issues, with a community leaderboard
- Weekly Nagar Report: Auto-generated, Gemini-written weekly summary of civic activity, concerns, and recommendations
- AI Civic Assistant: A conversational Gemini-powered chatbot answering citizen questions about civic data in natural language
- Anonymous Reporting: Citizens can report sensitive issues without their identity being publicly displayed
- Impact Dashboard: Visual analytics on resolution rates, issue trends, and zone-level performance

Technologies Used

React, Tailwind CSS, Node.js, JavaScript, HTML5, CSS3, React Router, Recharts, Lucide React, Leaflet.js

Google Technologies Utilized

- Google AI Studio (core development environment, as mandated)
- Google Gemini 1.5 Pro API (image analysis, duplicate detection, predictive insights, conversational assistant, automated reporting)
- Firebase Authentication
- Firebase Firestore (real-time database)
- Firebase Storage
- Firebase Hosting (deployment)